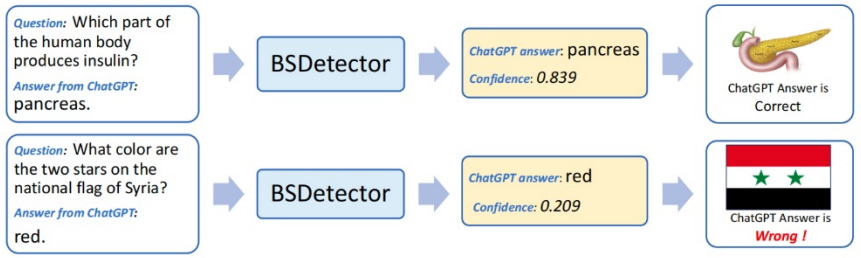
BSDetector

BSDetector[14]



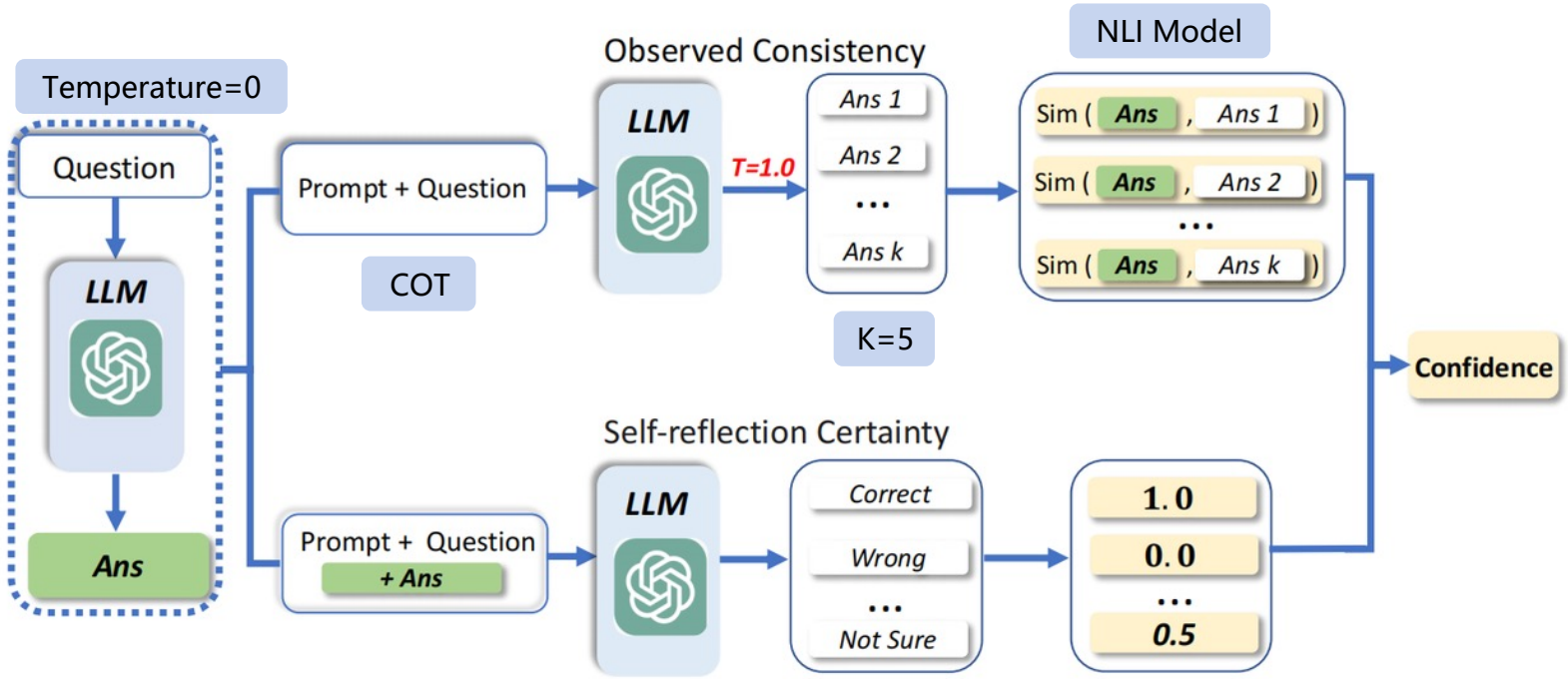
☐ Pipeline Overview

Two complementary estimators are combined into a single confidence score:

$$C = \beta O + (1 - \beta)S$$

where

- O: Observed Consistency (extrinsic)
- S: Self-reflection Certainty (intrinsic)
- $\beta = 0.7$



☐ Observed Consistency

$$s_i = \frac{1}{2}(1 - p_{contradiction} + 1 - p'_{contradiction}), i = 1,2,\dots,k$$

$$o_i = \alpha s_i + (1 - \alpha) r_i, \alpha = 0.8$$

$$O = o_i$$

☐ Self-reflection Certainty

"Is your previous answer correct?

A) Correct B) Incorrect C) I'm not sure."

Scores are assigned as

$$A = 1.0, B = 0.0, C = 0.5$$

and averaged over multiple follow-ups to form S.

[14] Quantifying Uncertainty in Answers from any Language Model and Enhancing their Trustworthiness, ACL 2024

BSDetector[14]

- **Task Type:** Open-ended generation

- **Introduction:** BSDetector combines extrinsic semantic similarity across sampled responses with intrinsic self-reflection scores from the LLM to compute a confidence estimate for any generated answer. It quantifies decoding uncertainty and provides a black-box procedure for assessing the reliability of LLM outputs without accessing internal probabilities or model weights.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Consensus-based	Decoding	Black-box	Post-hoc	Medium	Ranking

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